E4E Weekly Meeting

October 10, 2025

UC San Diego

Schedule

- General Announcements
- Social Media
- All Call Update
  1. FishSense
  2. Smartfin
  3. Active Learning
  4. Website
- Project Updates
  1. FishSense
  2. Hall Pass

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Announcements

- Servers SHOULD be back tonight
- Comet-ML Hosted on Fabricant?

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All Call Update

1. FishSense
2. Smartfin
3. Active Learning
4. Website

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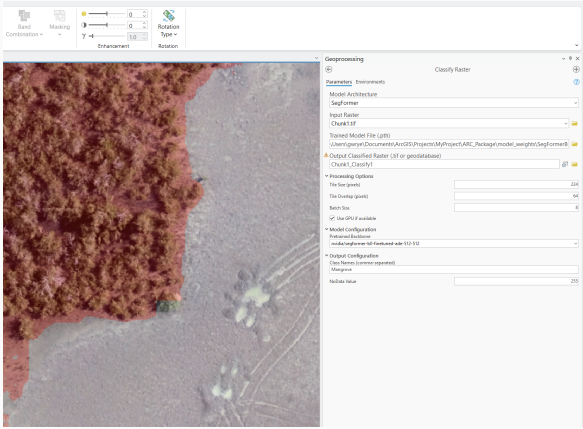
FishSense

Hall Pass

Mangrove Monitoring

ArcGIS Tool



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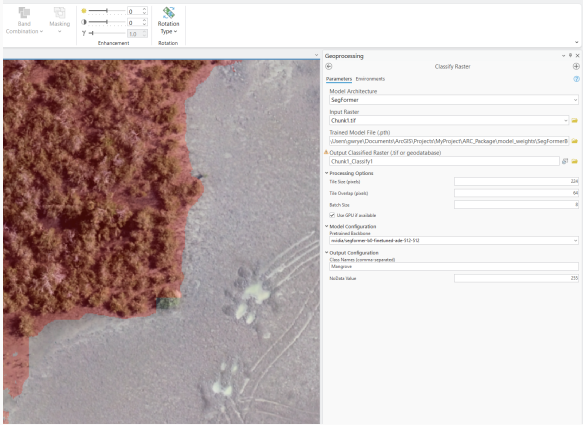
End-to-End ML Pipeline Completed

GeoTiff → Dataset Creation → Model Training → ArcGIS Deployment



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Small Data Processing Optimization

- Made optimization to GeoTiff $\rightarrow$ Dataset pipeline, improving asymptotic runtime from $O(n^2)$ to $O(n)$ for number of tiles n
- Stores temporary, batch-wise image/label files and collects everything at the end instead of modifying the memory mapped array periodically

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Next Steps

- Show SIO team the ArcGIS tool
- Get Feedback on models and tool
- Create human infrastructure identification model

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Thanks All!

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